DATA SCIENCE · MOCK INTERVIEW

Pack B — GreenWatt

Renewable Energy Forecasting

DOMAIN	PROBLEM TYPE	DURATION
Operations / Energy	Regression / Time Series	55 minutes total

⚠️ **INTERVIEWER GUIDE —** Do not share this document with the candidate before or during the session.

Interview Flow

Block	Your Role	Time	Cumulative
Welcome	Ask: *"Tell me about yourself — what brings you here?"*	2 min	2 min
Live Coding	Share problems. Watch screen. Ask comprehension question verbally.	15 min	17 min
Case Setup	Read the scenario aloud. Let them take notes.	3 min	20 min
Case Drive	They drive. Reveal info only when asked. Challenge their choices.	25 min	45 min
Synthesis	*"Can you summarize your recommendation?"*	3 min	48 min
Q&A	They ask you questions. Improvise as a senior DS consultant.	5 min	53 min
Debrief	Score the rubric together. One strength, one area to improve.	10 min	63 min

Case Scenario

Opening Script *(Read Aloud — 2 minutes)*

> *"Our client is **GreenWatt**, a renewable energy company that operates 50 onshore wind farms across Norway, Sweden, and Denmark. They sell electricity to the grid, but grid operators require day-ahead power production forecasts — and they penalize forecast errors.*
>
> *GreenWatt's current forecasting system is a basic statistical model built 5 years ago. It consistently over-predicts output, especially during low-wind periods. They're currently paying about **€8 million per year** in grid imbalance penalties.*
>
> *They've asked a consulting team to design a better forecasting system that reduces these penalty costs. They also have a strategic question: they're about to **acquire 20 new offshore wind farms**, and they need the forecasting system to work for those too."*

Information to Reveal — Only When Asked

Rule: Never volunteer information. If they don't ask, they don't get it.

If They Ask About…	What to Say
Current forecasting approach	Simple linear regression on weather forecasts from the national met service. Built by an engineer who left. Not retrained since 2020.
Data available	Hourly sensors per farm: wind speed, temperature, humidity, atmospheric pressure. 5 years of historical power output. Weather forecasts available 48 hours ahead, hourly granularity.
Power curve / non-linearity	**Show Exhibit 2.** Non-linear relationship — cut-in speed (turbines don't spin below it), cut-out speed (shut down for safety). Current linear model can't capture this — partly explains the systematic over-prediction.
Forecast accuracy	**Show Exhibit 1.** One-week sample: current forecast consistently overshoots, especially during dips. MAPE across all farms: 22%. Industry best practice: 10–12%.
Penalty structure	Proportional to absolute error. Over-prediction and under-prediction penalized equally. Rate: €30 per MWh of error.
The 50 farms	Vary in geography, turbine models, terrain. Some coastal, some inland. Performance characteristics differ.

The 20 new offshore farms	Offshore farms: higher and more consistent wind, but different turbulence. GreenWatt has **zero** operational data from these farms. They come online in 8 months.

Exhibits *(Screen-share only when asked)*

EXHIBIT 1 — Show when asked about forecast accuracy

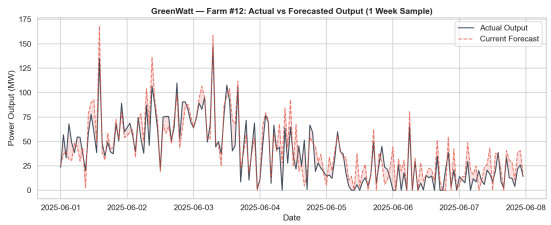
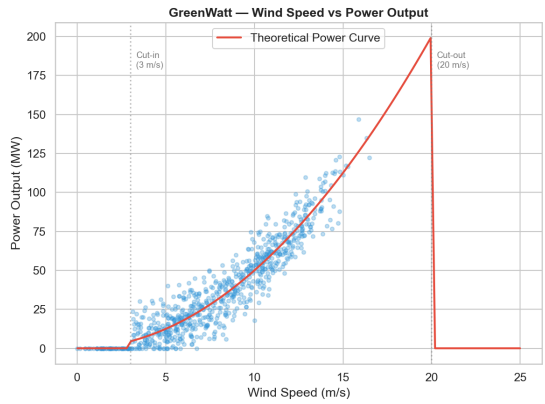


EXHIBIT 2 — Show when asked about the power curve



Challenges to Pose *(Pick 2–3 during Structure phase)*

On one model vs many

"You proposed [model]. But each farm has different characteristics. Are you training one model or 50? What are the tradeoffs?"

On non-linearity

"The power curve is clearly non-linear. How does your model capture that? Would a linear model work at all?"

On forecast uncertainty propagation

"Weather forecasts themselves have errors. Your model is predicting based on predicted wind speed, not actual wind speed. How does that uncertainty propagate through your system?"

On evaluation strategy

> *"How would you measure if your new model is actually saving them money? What's your evaluation strategy?"*

The Twist *(Drop around minute 18)*

DROP THIS AT 18:00

"Here's the strategic question: GreenWatt is acquiring 20 offshore wind farms that will come online in 8 months. There is zero historical production data from these farms. The turbine models are different from anything in the current fleet. How would you make your forecasting system work for them from day one?"

What a strong answer looks like: Transfer learning or domain adaptation from the 50 existing farms. Physics-based power curve baseline calibrated with manufacturer specs. As real data accumulates, shift to data-driven models. Hierarchical modeling: learn shared patterns across farms, then farm-specific fine-tuning. Could also use comparable offshore farms from public datasets as proxy.

What a Strong Candidate Covers

1. **Reframes clearly:** *"How do we reduce forecast errors from 22% MAPE to under 12%, cutting €8M in penalties, and scale to 20 new offshore farms with no historical data?"*
2. **Recognizes the non-linearity:** Power curve is not linear — this is the fundamental flaw in the current model.
3. **Smart model choice for time series:** Gradient boosting with lag features, LSTM for sequential patterns, or a physics-hybrid. Not just "use ARIMA."
4. **Feature engineering:** Lag features, rolling averages, weather forecast error as a feature, time-of-day/season, turbine-specific characteristics.
5. **One model vs many — takes a position:** Likely hierarchical — shared base model with farm-specific fine-tuning. Explains why.
6. **Cold start solution:** Physics-based baseline + transfer learning + manufacturer specs. Iterative improvement as data accumulates.
7. **Quantifies:** €8M ÷ 22% MAPE = €364K per percentage point. Reducing to 12% = **€3.6M annual savings**. The 20 new farms add another €4M penalty risk if not forecasted well.
8. **Proper evaluation:** Walk-forward validation (NOT random split for time series). Backtesting + A/B test on subset before full rollout.

Timing Guide

Phase	Target Time	Notes
Read scenario aloud	0:00 – 2:00	
Reframe + their questions	2:00 – 7:00	Show exhibits when relevant
Their structure	7:00 – 20:00	Challenge at 12:00 and 16:00
Drop the twist	18:00	The big one — cold start
Synthesis	22:00 – 25:00	*“Wrap up with a recommendation?”*

Coding Problems

Problem 1 — Wind Farm Performance Summary *(7 minutes)*

Dataset: `wind_production.csv` — columns: `timestamp, farm_id, wind_speed_ms, temperature_c, humidity_pct, pressure_hpa, power_output_mw, forecast_mw`

Note: Some values may be missing.

Write a Python function that returns a DataFrame with one row per **farm**, containing:

```
1. farm_id
2. avg_output — Average daily power output in MW (1 decimal)
3. avg_forecast — Average daily forecast in MW (1 decimal)
4. mape — Mean Absolute Percentage Error (2 decimal). Formula: mean(|actual -
   forecast| / actual) * 100. Exclude rows where actual = 0.
5. forecast_bias — Average of (forecast - actual). Positive = over-predicting. (1
   decimal)
6. data_completeness — % of rows with no missing values across all columns (1
   decimal)

→ Sort by mape descending (worst farms first). Show only the top 10.
```

Problem 2 — Wind Speed Binning & Power Curve Analysis *(7 minutes)*

Using the same dataset. Bin wind speed into:

```
• Calm: 0-3 m/s
• Light: 3-7 m/s
• Moderate: 7-12 m/s
• Strong: 12-20 m/s
• Extreme: >20 m/s

For each bin: count, avg_power (1 decimal), avg_forecast_error = avg |forecast - actual|
(1 decimal), pct_overpredict = % where forecast > actual (1 decimal).
Drop rows where wind_speed_ms is missing.
Bonus: In which bin does the model perform worst? Why?
```

Code Comprehension — Conversational *(no coding required)*

```python
def detect_anomalies(df, farm_id, window=24, threshold=2.5):
    farm = df[df['farm_id'] == farm_id].copy()
    farm['timestamp'] = pd.to_datetime(farm['timestamp'])
    farm = farm.sort_values('timestamp')
```

```
    farm['rolling_mean'] = farm['power_output_mw'].rolling(window, min_periods=1).mean()
    farm['rolling_std']  = farm['power_output_mw'].rolling(window, min_periods=1).std()

    farm['z_score']    = (farm['power_output_mw'] - farm['rolling_mean']) / farm['rolling_std']
    farm['is_anomaly'] = farm['z_score'].abs() > threshold

    return farm[farm['is_anomaly']][['timestamp', 'power_output_mw', 'rolling_mean', 'z_score']]
```

Ask these three questions verbally:

1. **"Walk me through what this function does."**
   - Expected: Detects anomalous power readings for one farm using a rolling Z-score. Calculates rolling mean and std over 24 periods, computes Z-score per reading, flags anything >2.5 std devs from the rolling mean.
2. **"What are the limitations of this approach for wind farm data?"**
   - Expected: Wind output is naturally variable — a sudden drop may just mean the wind stopped, not an equipment failure. The method doesn't account for wind speed as an input. A better approach would flag output as anomalous **given** current wind conditions, not just relative to recent output.
3. **"The `min_periods=1` parameter — what does it do and why might it cause problems?"**
   - Expected: Allows rolling computation with fewer than `window` observations (useful at start of series). But with very few periods, std is unreliable — could produce false anomalies at the beginning of the time series.

Interviewer Rules

→ **Screen sharing required** for coding. They can look up syntax — no AI tools.

→ **Never volunteer information.** If they don't ask, they don't get it.

→ **Challenge model choices.** "Why not a neural network?" "How would you evaluate this in production?"

→ **Drop the cold start twist** at 18 minutes.

→ **Use the scoring rubric** — fill it out during or right after.

→ **Keep it conversational.** You're a friendly colleague, not an examiner.

Make it real. The best gift you can give your friend is an honest, challenging mock. Push back. Don't go easy. Good luck! 🏔️